Electrocardiogram Signal-Based Cardiac Abnormality Detection

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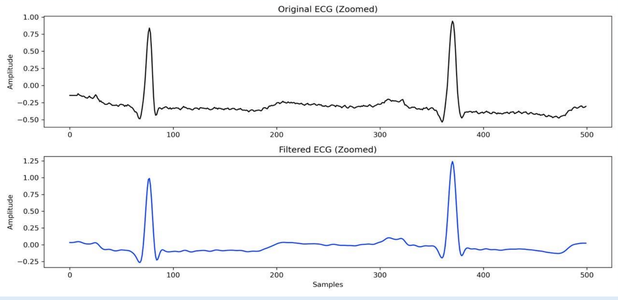
INTRODUCTION

- ECG signals are widely used for cardiac monitoring [1].
- Manual analysis is time-consuming and subjective [1].
- Signal noise and variability make detection challenging [1].
- Classical methods rely on handcrafted features.
- Deep learning models learn features automatically [3][4].
- Objective: compare SVM and CNN for ECG classification.

DATA PROCESSING

- Dataset: MIT-BIH Arrhythmia Database [2]
- ECG signals are time-series data.
- Signals contain noise and variability [2].
- Abnormal patterns are difficult to detect.
- Preprocessing improves signal quality:
  - Filtering
  - Normalization
  - Segmentation



METHODOLOGY

- Two approaches are compared for ECG classification.

- Traditional Method: Support Vector Machine
  - Feature extraction from ECG signals
  - Dimensionality reduction using PCA
  - Classification based on handcrafted features

- Deep Learning Method: Convolutional Neural Network
  - Automatic feature extraction [3][4]
  - Learns temporal patterns from raw signals [3][4]
  - End-to-end classification model [3][4]

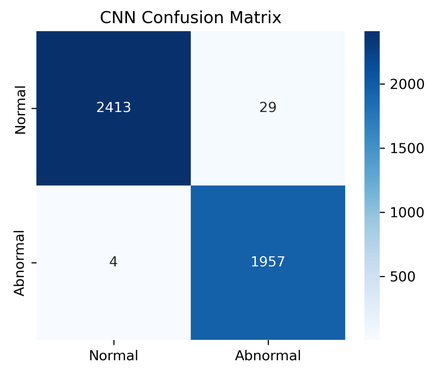


Figure 1. Confusion Matrix of CNN Model

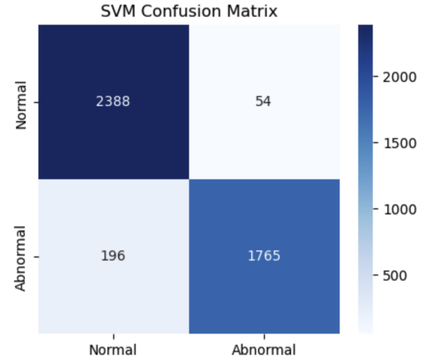


Figure 2. Confusion Matrix of SVM Model

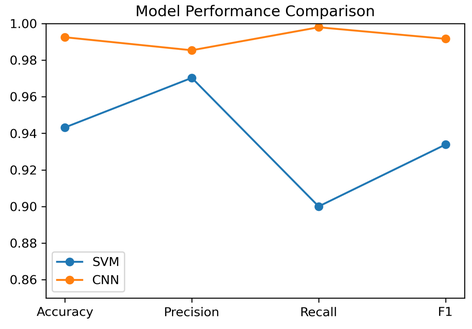


Figure 3. Performance Comparison between SVM and CNN

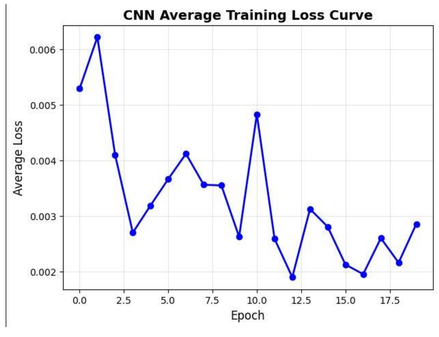


Figure 4. Training Loss Curve of CNN Model

- Figure 1–2 show that CNN produces fewer misclassifications than SVM.
- Figure 3 shows that CNN achieves consistently higher performance across all metrics.
- Figure 4 shows a decreasing training loss, indicating model convergence.

CONCLUSION

- This project compares a traditional SVM model and a deep learning CNN model for ECG signal classification.
- The results show that CNN consistently outperforms SVM across all evaluation metrics, including accuracy, precision, recall, and F1-score.
- In particular, CNN achieves a significantly higher recall, indicating a stronger ability to correctly identify abnormal heartbeats.
- The number of false negatives is reduced from 196 in SVM to only 4 in CNN, demonstrating a clear improvement in detecting critical abnormal cases.
- This reduction in missed abnormal signals is especially important in medical applications, where detection accuracy is crucial.
- Overall, the results indicate that CNN provides a more effective and reliable solution for ECG classification compared to traditional approaches.

REFERENCES

[1] Goldberger et al., Circulation, 2000.
[2] PhysioNet MIT-BIH Database, 2026.
[3] Kiranyaz et al., ICASSP, 2019.
[4] Acharya et al., Applied Intelligence, 2019.